

SHETH L.U.J AND SIR M.V COLLEGE

Artificial Intelligence

Practical No.4

Aim : Decision Tree

Code:

```
pre:py - C:\Users\gazala\OneDrive\Desktop\pre:py [2.13.2]
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import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Load the Iris dataset
df = pd.read_csv('Iris.csv')

# Display the first five records
print("First five records:")
print(df.head())

# Select input features and target variable
X = df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
y = df['Species']

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

# Create the Decision Tree model using Gini Index
model = DecisionTreeClassifier(
    criterion='gini',

# Train the Decision Tree model
model.fit(X_train, y_train)

# Predict the classes for test data
y_pred = model.predict(X_test)

# Calculate the accuracy of the model
accuracy = accuracy_score(y_test, y_pred)

print("\nDecision Tree Accuracy:", accuracy)
print("Accuracy Percentage:", accuracy * 100, "%")

# Display the confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display precision, recall and F1-score
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Visualize the confusion matrix using Matplotlib
plt.figure(figsize=(6, 5))

plt.imshow(cm)
plt.title("Decision Tree Confusion Matrix")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.colorbar()

classes = model.classes_
```

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```
plt.colorbar()

classes = model.classes_
plt.xticks(range(len(classes)), classes, rotation=30)
plt.yticks(range(len(classes)), classes)

# Display values inside the confusion matrix
for i in range(len(classes)):
    for j in range(len(classes)):
        plt.text(j, i, cm[i, j], ha='center', va='center')

plt.tight_layout()
plt.show()

# Visualize the generated Decision Tree
plt.figure(figsize=(16, 9))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=model.classes_,
    filled=True,
    rounded=True
)

plt.title("Generated Decision Tree")
plt.show()

# Display feature importance
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

class_names=model.classes_,
filled=True,
rounded=True
)

plt.title("Generated Decision Tree")
plt.show()

# Display feature importance
importance = pd.Series(
    model.feature_importances_,
    index=X.columns
).sort_values(ascending=False)

print("\nFeature Importance:")
print(importance)

# Visualize feature importance
plt.figure(figsize=(8, 5))

plt.bar(
    importance.index,
    importance.values
)

plt.xlabel("Features")
plt.ylabel("Importance")
plt.title("Feature Importance in Decision Tree")
plt.xticks(rotation=20)

plt.tight_layout()
plt.show()
```

Output:

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```
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>>>
===== RESTART: C:/Users/gazala/OneDrive/Desktop/pno3.py =====
First five records:
  Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm  Species
0  1             5.1             3.5             1.4             0.2  Iris-setosa
1  2             4.9             3.0             1.4             0.2  Iris-setosa
2  3             4.7             3.2             1.3             0.2  Iris-setosa
3  4             4.6             3.1             1.5             0.2  Iris-setosa
4  5             5.0             3.6             1.4             0.2  Iris-setosa

Training records: 120
Testing records: 30

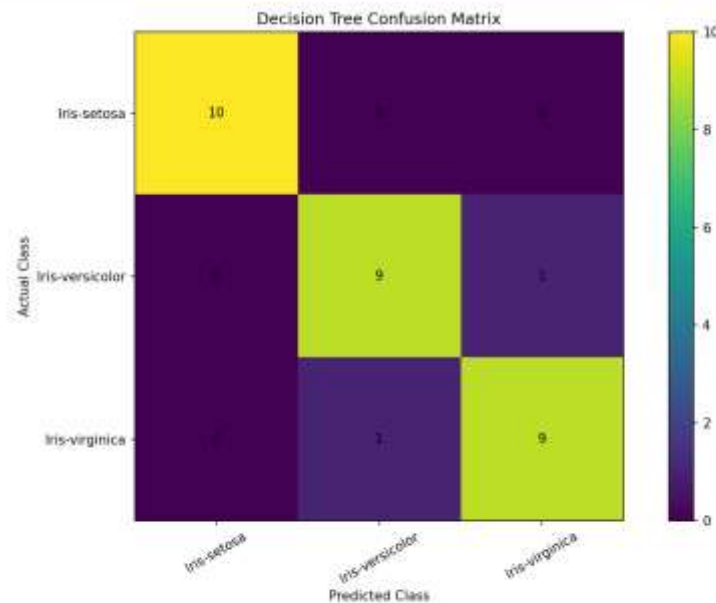
Decision Tree Accuracy: 0.9333333333333333
Accuracy Percentage: 93.33333333333333 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  1  9]]

Classification Report:
              precision    recall  f1-score   support

 Iris-setosa      1.00      1.00      1.00        10
 Iris-versicolor  0.90      0.90      0.90        10
 Iris-virginica   0.90      0.90      0.90        10

 accuracy          0.93      0.93      0.93        30
 macro avg         0.93      0.93      0.93        30
 weighted avg      0.93      0.93      0.93        30
```



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